

Double circulation: engineer the route

Teacher guide and worked answers

Explain how heart structure supports double circulation and calculate cardiac output from supplied data.

40 minutes. Use Lesson.html for interactive teaching or Lesson.pptx for editable slides. Slides.pdf is the static alternative. Select one pupil response route and print the shared evidence it requires. Detailed answers below are for staff.

Scheme of work

[_passsl/inputs/LAUNCH KS4 - 2026-27.xlsx](#) | LAUNCH Weekly - Summer | C32

Describe the heart & circulation.

Directly develops C32 through Pearson 8.7, 8.8 and 8.12. This LAUNCH exemplar adds named chambers, major vessels, pressure reasoning and calculations; it does not claim a wholly untouched topic or complete cardiovascular teaching.

Prior learning

Blood carries materials between organs.

Lungs exchange oxygen and carbon dioxide with blood.

A rate includes an amount and a time.

Success evidence

Trace the body-heart-lungs-heart-body route with named chambers and vessels.

Explain valves and the thicker left-ventricle wall using function.

Calculate output with correct units and state a limitation of the constructed data.

Equipment

Shared evidence sheet, one selected response route, pen and ruler.

Projector or offline HTML/PowerPoint; calculator where useful.

Preparation

Print the shared page, one selected response route and the exit page. Routes are alternatives, not extra tasks.

Read the worked answers and note any constructed data before teaching.

Practical care

Paper and screen lesson; no human sampling, exercise test, breath holding, medicine handling or personal health disclosure.

Ready to teach alternative

All evidence is supplied. Use paper diagrams and cards, pointing, annotation or dictation if a device is unavailable.

Teaching sequence

Slide	Stage	Minutes	Focus
1	Opening	0-2	Can you design a route with no wrong turn?
2	Retrieval	2-5	Direction, exchange and rate
3	I do	5-9	Trace the full double circuit
4	I do	9-12	Explain structure using its job
5	I do	12-15	Separate the pump from the valve
6	We do	15-18	Route engineering desk
7	We do	18-20	Pump data challenge
8	We do	20-22	Repair the wall claim
9	Hinge	22-25	Which definition survives the lung route?
10	You do	25-26	Produce a route-and-pump brief
11	You do	26-36	Make every arrow explain something
12	Review	36-38	Audience checks the exception
13	Review	38-39	A route and a calculation need units
14	Exit	39-40	Three useful checks
15	Exit	Within stage	Next useful step

Times are a suggested 40-minute route. Slides marked Within stage share the preceding stage allocation. The optional HTML timer starts only when selected.

1 Opening Can you design a route with no wrong turn?

Use a finger trace or token. Explain that the diagram is a map of flow, not a scale picture of chest anatomy.

2 Retrieval Direction, exchange and rate

R1 lungs. R2 amount and time. R3 no; pressure and volume are different quantities. Accept everyday pump analogies with limits.

3 I do Trace the full double circuit

Think aloud around the loop, naming pulmonary artery and pulmonary veins as the oxygen-rule exceptions. “Away/towards” defines the vessel type, not oxygen content.

4 I do Explain structure using its job

Trace atrium → ventricle on each side. A thicker left-ventricle wall supports greater pressure for the systemic circuit; do not claim it normally pumps a larger long-term volume than the right side.

5 I do Separate the pump from the valve

Use forward and reverse pressure panels. Model output separately: $75 \text{ beats/min} \times 60 \text{ mL/beat} = 4500 \text{ mL/min} = 4.5 \text{ L/min}$. These are constructed examples, not a target for pupils.

6 We do Route engineering desk

Body → vena cava → RA → RV → pulmonary artery → lungs → pulmonary veins → LA → LV → aorta → body. Accept abbreviations only after names are understood.

7 We do Pump data challenge

P: $80 \times 55 = 4400 \text{ mL/min} = 4.4 \text{ L/min}$. Q: $100 \times 70 = 7000 \text{ mL/min} = 7.0 \text{ L/min}$. Difference 2600 mL/min. No exercise test or individual health judgement is made.

8 We do Repair the wall claim

The thicker wall supports higher pressure for the body circuit. In stable circulation the two sides have matching average outputs, otherwise blood would accumulate in one circuit.

Reveal: Thicker muscle supports the higher pressure needed for the systemic route; it does not imply a permanently larger output.

9 Hinge Which definition survives the lung route?

First option correct. Re-trace pulmonary vessels if pupils classify by oxygen content.

A. The pulmonary artery is an artery because it carries blood away from the heart. - Direction defines an artery, even when its blood is relatively oxygen-poor.

B. Every vein must carry relatively oxygen-poor blood. - Pulmonary veins return relatively oxygen-rich blood from lungs to heart.

C. Valves supply the force that pumps blood. - Heart muscle creates pressure; valves help direct flow and prevent backflow.

10 You do Produce a route-and-pump brief

Use shared route/data page. Pointing with accurate spoken names is a valid access route.

11 You do Make every arrow explain something

Supported S1-S4; standard M1-M4; stretch T1-T4. If stuck, start from lungs → left atrium and build around the route.

12 Review Audience checks the exception

Pupil corrects a definition or defends an accurate away/towards explanation. Quiet teacher/TA audience is sufficient.

13 Review A route and a calculation need units

Model 4.4 L/min as an output, not heart rate. Ask pupils to add one missing unit or destination.

14 Exit Three useful checks

E1 right ventricle. E2 greater pressure for systemic circuit. E3 $5400 \text{ mL/min} = 5.4 \text{ L/min}$. Use errors to plan route versus rate re-teaching.

15 Exit Next useful step

Shared exit minute. No comparison with a pupil's pulse or health is required.

Answers and assessment

Retrieval and We do

R1 lungs; R2 amount/time; R3 pressure differs from volume. W1 full route: body → vena cava → right atrium → right ventricle → pulmonary artery → lungs → pulmonary veins → left atrium → left

ventricle → aorta → body. W2 $P = 4400 \text{ mL/min}$; $Q = 7000 \text{ mL/min}$. W3 thicker left wall supports higher pressure, not permanently larger output.

Hinge

Pulmonary artery carries blood away from heart, so it is an artery. Pulmonary veins carry relatively oxygen-rich blood towards heart. Muscle supplies pumping pressure; valves direct flow.

S1 / M1

Use the full route above. Septum separates left and right sides, limiting mixing of the two streams in the normal adult heart.

S2

An artery is defined by blood travelling away from the heart, not its oxygen content. The pulmonary artery runs from right ventricle towards lungs.

S3 / M3

P: $80 \times 55 = 4400 \text{ mL/min} = 4.4 \text{ L/min}$. Q: $100 \times 70 = 7000 \text{ mL/min} = 7.0 \text{ L/min}$. Difference $2600 \text{ mL/min} = 2.6 \text{ L/min}$.

S4 / M2

Pressure differences open/close valve leaflets, preventing backflow. The left ventricle has a thicker muscular wall to produce the higher pressure for systemic circulation. In stable circulation average outputs of the two sides match.

M4 / T1

Need stroke volume too. R: $120 \times 35 = 4200 \text{ mL/min}$; S: $70 \times 70 = 4900 \text{ mL/min}$. S has lower heart rate but greater output because its stroke volume is much larger.

T2

Some moved blood returns backwards rather than progressing through the circuit; useful net forward flow can be lower than gross chamber movement.

T3

Because the circuits connect, persistently unequal average outputs would cause blood to accumulate in one part and deplete another. The thicker wall is primarily a pressure adaptation.

T4

Output is a volume per time, so 5400 mL/min (if those were the measured volume data), not beats/min. Heart rate uses beats/min; a unit alone cannot diagnose health.

Exit and assessment

E1 right ventricle. E2 higher systemic pressure. E3 $60 \times 90 = 5400 \text{ mL/min} = 5.4 \text{ L/min}$. Secure core: route direction, pressure/valve mechanism and valid output units. Re-teach using away/towards if pulmonary vessels are misclassified.

Teaching and assessment notes

40-minute lesson. Zero-minute slides share the preceding stage allocation.

Choose one response route. Accept accurate speech, pointing with explanation, annotations or writing; record the support used.

A teacher or TA can be a quiet audience: ask one evidence question, then let the pupil revise or justify a decision.

All pump data are constructed. This lesson requires no pulse-taking, exertion, health comparison or physiological threshold.

Interactive route/valve/data decisions use the same evidence as W1-W3. Manual progression and a paper finger-trace provide equivalent reasoning.

The heart diagrams are simplified flow/structure models with text labels as well as colour; they are not diagnostic anatomical images.

Access and responsive teaching

Supported

Same core concepts with chunked prompts and read-aloud access. Record support used; a pathway is not a diagnosis or fixed attainment label.

Standard

Explain the supplied evidence independently using accurate scientific terms and units.

Stretch

Optional deeper transfer or evaluation. Higher-tier-only material is labelled explicitly and is not required for the core outcome.

Regulation

Offer solo or partner work, a quiet response, a pass and brief reset with clear re-entry. No forced disclosure or public performance.

Misconceptions to check

An artery is defined as carrying oxygen-rich blood.

Arteries carry blood away from the heart; the pulmonary artery carries oxygen-poor blood to the lungs.

Check: Ask what makes the pulmonary artery an artery.

The left ventricle pumps more blood per beat because its wall is thicker.

In steady circulation the two sides have matching average outputs; the thicker wall generates the higher pressure needed for the systemic route.

Check: Separate pressure from volume in the explanation.

Valves actively pump the blood.

Muscle contraction creates pressure; valves open/close with pressure differences to prevent backflow.

Check: Ask which structure supplies the pushing action.

A fast heart rate alone proves greater output.

Stroke volume matters too; output is the product of the two quantities.

Check: Compare both columns before judging a pump record.

Word	Meaning
artery	A vessel carrying blood away from the heart.
vein	A vessel carrying blood towards the heart.
atrium	A heart chamber that receives returning blood.
ventricle	A heart chamber that pumps blood out of the heart.
stroke volume	The volume pumped by a ventricle in one beat.
cardiac output	The volume pumped by a ventricle per minute: stroke volume multiplied by heart rate.

Scientific references

Pearson Edexcel GCSE Biology 1BI0 specification

Used to check relevant Foundation curriculum content; this is teacher-authored practice, not an official assessment or complete unit. Checked 9 September 2026.

NHLBI: How blood flows through the heart

Checked circulation route, major vessels and valves. Constructed pump values are original. Checked 9 September 2026.

NHLBI: What the heart looks like

Checked chamber roles, septum and muscular structure. All diagrams here are original simplified teaching models. Checked 9 September 2026.